

Pallavi Palecharu Venkatakrishna

Dallas, Texas

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 github.com/pallavipalechar  pallavi-pv/portfolio

Professional Summary

Full-stack software developer with 3+ years of experience in enterprise PLM systems, specializing in Siemens Teamcenter and web product development. Skilled in building user-focused web applications with React.js, Next.js, and REST APIs, with a focus on performance and reusable component design. Hands-on volunteer experience delivering complete full-stack solutions. Recently expanded skill set into Generative AI, prompt engineering, and DevOps practices. Authorized to work full-time in the United States under H-4 EAD. Exploring career opportunities in Siemens Teamcenter, Front-end, Full-stack or AI-focused developer roles.

Experience

Freelance Developer

January 2025 – Present

Pro bono

Dallas, USA

- Building and deploying client websites using Next.js, React.js, and Tailwind CSS with a focus on performance, SEO, and accessibility, shipping through Vercel.
- Managed client relationships and project scoping across multiple concurrent engagements, independently owning timelines and stakeholder communication in the absence of a dedicated project manager.

Frontend Developer and Product Designer

June 2024 – December 2024

EZ Technologies (Pro bono)

Mangalore, India

- Implemented high-performance web applications using Python and PHP for backend development, and React.js with JavaScript (ES6+) for frontend development. This improved mobile responsiveness by 30% and ensured cross-browser compatibility across Chrome, Firefox, and Safari.
- Collaborated with stakeholders to gather business requirements and translate them into detailed application workflows, process flows, and database designs, providing the foundation for end-to-end web application development.

Teamcenter Developer (PLM)

October 2022 – January 2024

Tata Consultancy Services (TCS)

Bangalore, India

- Delivered enterprise PLM solutions on Siemens Teamcenter14.2 for a large manufacturing client, configuring BMIDE customizations, Active Workspace6.2 UI component customizations, and developing .NET REST APIs to automate manual administrative workflows across 2-tier and 4-tier architectures.
- Authored technical onboarding documentation and led knowledge-transfer sessions to standardize Teamcenter customization practices and reduce new engineer ramp-up time, while providing ongoing Teamcenter PLM support.

Projects

IronGuard Chat - AI Safety Chatbot

 Repository

May 2026

- Developed a full-stack Retrieval-Augmented Generation chatbot with a React.js frontend, Flask backend, and a Hugging Face model underneath, with a prompt risk-analysis filter that catches harmful inputs before they ever reach the model.
- Moved to async processing and cut end-to-end response latency by about 70%, then layered in safety guardrails drawn from published AI safety research.

Kalpavedike - Religious Organization Website

 Repository

December 2025

- Designed and deployed a fully functional, accessible website for a devotee community, achieving 30% faster load times through Next.js server-side rendering and image optimization.
- Ensured seamless, responsive UX across all device sizes using Tailwind CSS utility-first design system.

Technical Skills

Technologies/Frameworks: React.js, Next.js, Tailwind CSS, Bootstrap, Redux

Programming Languages: Python, JavaScript (ES6+), Java, C, C++, .NET, HTML5, CSS3, PHP

Backend & APIs: Node.js, RESTful APIs, Flask

Databases and Enterprise Platforms: MongoDB, MySQL, Siemens Teamcenter, BMIDE, Active Workspace, ERP-Next

AI / ML: OpenAI API, Hugging Face Transformers, GPT-2, LangChain, Prompt Engineering, Generative AI

Tools and DevOps: Git, GitHub, GitHub Actions, Vercel, Postman, Figma, Visual Studio Code, Jira, WordPress

Concepts: Web/UI Development, Component Architecture, CI/CD, Wireframing, Prototyping, Agile/Scrum, Debugging

Education

Master of Computer Applications (MCA)

2020 – 2022

Srinivas Institute of Technology, affiliated with Visvesvaraya Technological University (VTU)

Karnataka, India

Certifications

- **AI Fluency Framework and Foundations:** AI Concepts and Responsible Use – **Anthropic**
- **Claude 101:** Prompt Engineering and Applied AI with Claude – **Anthropic**
- **AI Whisperer :** Crafting Effective Prompts - Generative AI, Prompt Engineering – **University of South Florida**
- **AI and Career Empowerment :** AI Fundamentals, Career Development – **University of Maryland**
- **UI/UX Design :** User Interface Design, User Experience – **Great Learning**
- **UX Foundations :** Interaction Design, Interaction Design Principles – **LinkedIn Learning**